82bf030f-79fa-44fd-a6c3-542ff059b18c

<prev_user>

<request>

You are working on an issue in the 'mojohaus/xml-maven-plugin' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/xml-maven-plugin/xml-maven-plugin (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

Consider the following problem statement:

<problem_statement>

This section details on the original issue you should resolve

<issue_title>Does not follow redirects when resolving schema for catalog.xml</issue_title>
<issue_description>I suddenly got the following error when executing </issue_description>`validate` in build which has previously worked for many years:
```
[ERROR] Failed to execute goal org.codehaus.mojo:xml-maven-plugin:1.0.2:validate (default) on project X: While parsing projectroot/catalog.xml, at http://www.oasis-open.org/committees/entity/release/1.1/catalog.xsd, line 2,  column 35: error: s4s-elt-character: Non-whitespace characters are not allowed in schema elements other than 'xs:appinfo' and 'xs:documentation'. Saw '301 Moved Permanently'.
[ERROR] While parsing projectroot/catalog.xml, at http://www.oasis-open.org/committees/entity/release/1.1/catalog.xsd, line 4,  column 36: error: s4s-elt-character: Non-whitespace characters are not allowed in schema elements other than
```

'xs:appinfo' and 'xs:documentation'. Saw '301 Moved Permanently'.
[ERROR] While parsing projectroot/catalog.xml, at http://www.oasis-open.org/committees/entity/release/1.1/catalog.xsd, line 5, column 20: error: s4s-elt-character: Non-whitespace characters are not allowed in schema elements other than 'xs:appinfo' and 'xs:documentation'. Saw 'nginx'.
[ERROR] While parsing projectroot/catalog.xml, at http://www.oasis-open.org/committees/entity/release/1.1/catalog.xsd, line 6, column 3: fatal error: The element type "hr" must be terminated by the matching end-tag "".
[ERROR] While parsing projectroot/catalog.xml, at http://www.oasis-open.org/committees/entity/release/1.1/catalog.xsd, line 6, column 3: fatal error: The element type "hr" must be terminated by the matching end-tag "".
```

This hints about some error related to redirecting: "Saw '301 Moved Permanently'".
Executing `curl -sLD - http://www.oasis-open.org/committees/entity/release/1.1/catalog.xsd` confirms the schema-URL has indeed been moved:
```bash
$ curl -sLD - http://www.oasis-open.org/committees/entity/release/1.1/catalog.xsd
HTTP/1.1 301 Moved Permanently
Server: nginx
Date: Wed, 11 Nov 2020 18:31:18 GMT
Content-Type: text/html
Content-Length: 162
Connection: keep-alive
Location: https://www.oasis-open.org/committees/entity/release/1.1/catalog.xsd

HTTP/1.1 200 OK
Server: nginx
Date: Wed, 11 Nov 2020 18:31:19 GMT
Content-Type: application/xml
Content-Length: 8744
Connection: keep-alive
Last-Modified: Wed, 11 Nov 2020 03:21:33 GMT
ETag: "17c9e0-2228-5b3cc4ac9e940"
Accept-Ranges: bytes
Strict-Transport-Security: "max-age=31536000; includeSubDomains" always
...
```
```

I have changed the URL in my `catalog.xml` to https, and this solves the problem, but the http-client used internally by xml-maven-plugin should maybe be set up to follow redirects?

Comments on the Issue (you are @copilot in this section)

<comments>
<comment_new><author>@jochenw</author>
Just for the record: It appears, that the http URL works again, so we don't have something reproducable.
</comment_new>
<comment_new><author>@jochenw</author>
@runeflobakk Could you provide the relevant snippet from your POM? Catalog files? Whatever else you are using?
</comment_new>
</comments>

</problem_statement>

Implement the necessary changes to the repository so that the requirements specified in the <problem_statement> are met.

Ensure that you lint, build and test early and iteratively to validate each code change thoroughly.

<prev_assistant>

Sorry, you've hit a rate limit that restricts the number of Copilot model requests you can make within a specific time period. Please try again in 46 minutes. If the problem persists, please contact GitHub Support, including the request ID `8402:35FCAE:144518B:1F4680B:68E2B7D3`. To retry, leave a comment on this pull request asking Copilot to try again.

<current>

<request>

You have been given comments on the previous commits you made in the repository.

You are working on an issue in the 'mojohaus/xml-maven-plugin' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/xml-maven-plugin/xml-maven-plugin (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

Consider the following problem statement:

<problem_statement>

This section includes details on the pull request with code changes you have already submitted to fix the problem above.

<pr_title>
[WIP] Does not follow redirects when resolving schema for catalog.xml
</pr_title>

<pr_description>
- [x] Understand the issue - plugin doesn't follow HTTP redirects when resolving schemas
- [x] Identify the root cause - URL.openStream() doesn't follow HTTP->HTTPS redirects
- [x] Create helper method to open URL connections with redirect following enabled
- [x] Update resolveAsURL method to use the helper method
- [x] Update asInputSource method to use the helper method
- [x] Build and test changes
- [ ] Add documentation for redirect behavior (optional)
- [ ] Manual testing with actual redirect URLs (optional)

<details>

<summary>Original prompt</summary>

>
> ----

```
&gt;
&gt; *This section details on the original issue you should
resolve*
&gt;
&gt; <issue_title>Does not follow redirects when resolving
schema for catalog.xml</issue_title>
&gt; <issue_description>I suddenly got the following error when
executing </issue_description></details></
pr_description>`validate` in build which has previously worked
for many years:
> ```
> [ERROR] Failed to execute goal org.codehaus.mojo:xml-maven-
plugin:1.0.2:validate (default) on project X: While parsing
projectroot/catalog.xml, at http://www.oasis-open.org/
committees/entity/release/1.1/catalog.xsd, line 2, column 35:
error: s4s-elt-character: Non-whitespace characters are not
allowed in schema elements other than 'xs:appinfo' and
'xs:documentation'. Saw '301 Moved Permanently'.
> [ERROR] While parsing projectroot/catalog.xml, at http://
www.oasis-open.org/committees/entity/release/1.1/catalog.xsd,
line 4, column 36: error: s4s-elt-character: Non-whitespace
characters are not allowed in schema elements other than
'xs:appinfo' and 'xs:documentation'. Saw '301 Moved
Permanently'.
> [ERROR] While parsing projectroot/catalog.xml, at http://
www.oasis-open.org/committees/entity/release/1.1/catalog.xsd,
line 5, column 20: error: s4s-elt-character: Non-whitespace
characters are not allowed in schema elements other than
'xs:appinfo' and 'xs:documentation'. Saw 'nginx'.
> [ERROR] While parsing projectroot/catalog.xml, at http://
www.oasis-open.org/committees/entity/release/1.1/catalog.xsd,
line 6, column 3: fatal error: The element type "hr" must be
terminated by the matching end-tag "".
> [ERROR] While parsing projectroot/catalog.xml, at http://
www.oasis-open.org/committees/entity/release/1.1/catalog.xsd,
line 6, column 3: fatal error: The element type "hr" must be
terminated by the matching end-tag "".
>```
>
> This hints about some error related to redirecting: "Saw '301
Moved Permanently'".
> Executing `curl -sLD - http://www.oasis-open.org/committees/
entity/release/1.1/catalog.xsd` confirms the schema-URL has
indeed been moved:
> ```bash
> $ curl -sLD - http://www.oasis-open.org/committees/entity/
release/1.1/catalog.xsd
```

```
> HTTP/1.1 301 Moved Permanently
> Server: nginx
> Date: Wed, 11 Nov 2020 18:31:18 GMT
> Content-Type: text/html
> Content-Length: 162
> Connection: keep-alive
> Location: https://www.oasis-open.org/committees/entity/
release/1.1/catalog.xsd
>
> HTTP/1.1 200 OK
> Server: nginx
> Date: Wed, 11 Nov 2020 18:31:19 GMT
> Content-Type: application/xml
> Content-Length: 8744
> Connection: keep-alive
> Last-Modified: Wed, 11 Nov 2020 03:21:33 GMT
> ETag: "17c9e0-2228-5b3cc4ac9e940"
> Accept-Ranges: bytes
> Strict-Transport-Security: "max-age=31536000;
includeSubDomains" always
> ...
> ```
>
> I have changed the URL in my `catalog.xml` to https, and this
solves the problem, but the http-client used internally by xml-
maven-plugin should maybe be set up to follow redirects?
>
> ## Comments on the Issue (you are @copilot in this section)
>
><comments>
> <comment_new><author>@jochenw</author>
> Just for the record: It appears, that the http URL works
again, so we don't have something reproducable.
> </comment_new>
> <comment_new><author>@jochenw</author>
> @runeflobakk Could you provide the relevant snippet from
your POM? Catalog files? Whatever else you are using?
> </comment_new>
> </comments>
>

Fixes mojohaus/xml-maven-plugin#59

Comments on the PR (you are @copilot in this section)
```

<comments>

<pr_comments>

<comment_new>
<comment_id>3369275612</comment_id>
<author>@slachiewicz</author>
@copilot continue working on fix
</comment_new>

</pr_comments>

</comments>

The last **2** git commits in this branch are the changes you have made so far. Use those as your change commit history.
</problem_statement>

Analyze and determine if action or explanation is needed
<rules_for_making_changes>
* Start by fully understanding <problem_statement> and the <comments> before making any changes. You will only address the <comment_new> comments. Reply to new comments as you address them. Ignore comments that are not actionable or directed at you.
* For each new comment, determine:
    - If it a request for changes, a question on the code or a suggestion.
    - Is it addressed to you or someone else, e.g. "Hey @username, ..." or "@username can you ...".
    - Is it a general comment or praise.
* If there are no comments you need to address or reply to, you can stop now without exploring or explaining yourself.
* If the user's ask is not clear, you can reply to the comment and ask for clarification.
* If an action or explanation is needed, follow the steps below.
</rules_for_making_changes>

Ensure that you lint, build and test early and iteratively to validate each code change thoroughly.